

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Questions

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks 50

Duration :60 Minutes

Annexure A
Scenario-Based Questions

Code of Conduct:

I P Jyothika / 192311156 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V. Sreekanth

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

1. Low Contrast Image

Scenario Statement: A grayscale image appears dull with very little difference between object and background intensities.

(a) Clearly interpret the scenario and explain the cause of low contrast. In digital image processing, image contrast is defined as the relative difference in luminance or grey-level intensity between features and background within the same field of view. In this given scenario, the grayscale image exhibits poor dynamic range, meaning that the pixel intensities are severely compressed into a very narrow band of grey levels (for example, concentrated tightly between intensity values 80 and 120 on a standard 8-bit scale of [0, 255]). This produces a narrow, highly clustered intensity histogram lacking deep blacks (near 0) or bright whites (near 255).

- **Inadequate or Uniform Illumination:** Low ambient lighting, overcast atmospheric conditions, or highly diffuse background illumination that reduces specular reflection from object surfaces during capture.
- **Imaging Sensor & Optical Limitations:** Sensors with poor dynamic range, low signal-to-noise ratio (SNR), incorrect camera aperture settings, or improper exposure times that fail to capture the full radiometric range.
- **Environmental Factors:** Atmospheric haze, fog, or underwater scattering that attenuates light rays before reaching the optical lens.

(b) Identify all key issues affecting image visibility.

- **Object-Background Merging & Loss of Contour Definition:** Because foreground target intensities closely match background intensities, boundary gradients are near zero. As a result, object outlines blur into the background, making target identification visually impossible.
- **Information Masking & Feature Suppression:** Fine structural details, surface textures, and subtle edges are completely masked within the compressed intensity cluster. Downstream automated computer vision algorithms (such as Sobel edge detection, Canny edge detection, or Otsu threshold segmentation) fail due to insufficient gradient strength.
- **Underutilized Dynamic Range:** Large portions of the display intensity spectrum (e.g., 0–79 and 121–255) contain zero pixels, rendering the display capability wasted and resulting in a visually dull, washed-out image.

(c) Apply an appropriate enhancement technique to address the problem. To solve the low contrast problem effectively, we apply Histogram Equalization (or Contrast Limited Adaptive Histogram Equalization - CLAHE for localized enhancement). Histogram Equalization is a non-linear spatial domain point-processing technique that maps input grey levels r_k into output grey levels s_k such that the output probability density function (PDF) approaches a uniform distribution across $[0, L-1]$.

- **Normalized Probability Density Function (PDF):** $p_r(r_k) = n_k / N$, where n_k is the number of pixels with intensity r_k , and N is the total pixel count in the image.
- **Cumulative Distribution Function (CDF):** $CDF(r_k) = \sum [p_r(r_j)]$ for $j = 0$ to k .

- **Transformation Formula:** $s_k = T(r_k) = \text{round}((L - 1) * \text{CDF}(r_k)) = \text{round}(255 * (\text{CDF}(r_k) / N))$ (for $L = 256$ grey levels).

(d) Justify your choice with logical reasoning. Histogram Equalization is chosen because it is an automated, parameter-free technique that calculates the optimal transformation directly from the statistical distribution of the input image. Unlike simple linear contrast stretching (which requires manual setting of upper and lower clipping bounds), histogram equalization automatically expands tight intensity clusters proportional to their pixel frequency. It maximizes the information entropy of the image, ensuring that every display grey level carries a balanced amount of visual information.

(e) Present your explanation in a clear and structured manner. By transforming the compressed intensity distribution using the image's cumulative distribution function (CDF), dark background pixels are mapped closer to lower intensities (towards 0), while bright object pixels are stretched toward higher intensities (towards 255). This expands the narrow histogram across the full $[0, 255]$ spectrum, boosting global contrast, establishing crisp object-background boundaries, and restoring hidden surface details.

2. Uneven Illumination

Scenario Statement: An image captured under non-uniform lighting shows bright and dark patches.

(a) Interpret the scenario and explain the impact of uneven illumination. In this scenario, the image suffers from low-frequency spatial illumination gradients across its field of view, causing artificial bright spots (glare/over-exposure) in well-lit regions and deep shadows (under-exposure) in shadowed regions. According to the foundational image formation model, an image $f(x,y)$ is represented as the scalar product of two components: $f(x,y) = i(x,y) * r(x,y)$, where $i(x,y)$ represents the illumination component (determined by the light source) and $r(x,y)$ represents the reflectance component (determined by the physical surface properties of the objects). Uneven illumination means $i(x,y)$ varies significantly across spatial coordinates (x,y) , corrupting the observed intensities and causing identical physical materials to exhibit completely different pixel values across the image.

(b) Identify key factors causing intensity variation.

- **Directional & Single-Source Lighting:** Unshielded point light sources or harsh sunlight casting strong directional shadows and creating extreme specular glare on facing surfaces.
- **Optical Vignetting & Lens Distortion:** Light attenuation occurring toward the periphery of optical camera lenses, resulting in central brightness and darker image corners.
- **3D Surface Geometry & Occlusion:** Complex non-planar objects blocking ambient light from reaching recessed surfaces, creating localized dark patches.

(c) Apply a suitable enhancement method to normalize illumination. To eliminate non-uniform lighting while preserving object details, we apply Homomorphic Filtering in the frequency domain (or Top-Hat Morphological Transformations / Adaptive Thresholding). Homomorphic filtering relies on the fact that illumination $i(x,y)$ consists of low spatial frequencies (slow variations across space), whereas reflectance $r(x,y)$ consists of high spatial frequencies (sharp object edges and fine details). It separates the multiplicative components into additive components using logarithmic transformation:

- **Step 1 (Logarithmic Transformation):** $\ln[f(x,y)] = \ln[i(x,y) * r(x,y)] = \ln[i(x,y)] + \ln[r(x,y)]$
- **Step 2 (Frequency Domain Conversion):** $F\{\ln[f(x,y)]\} = F\{\ln[i(x,y)]\} + F\{\ln[r(x,y)]\} \implies Z(u,v) = F_i(u,v) + F_r(u,v)$
- **Step 3 (High-Pass / Modified Filtering):** Apply Homomorphic Filter $H(u,v)$ where $H(u,v)$ attenuates low frequencies ($\gamma_L < 1$) and amplifies high frequencies ($\gamma_H > 1$):
 $S(u,v) = H(u,v) * Z(u,v)$
- **Step 4 (Inverse Fourier & Exponential):** $s(x,y) = F^{-1}\{S(u,v)\} \implies g(x,y) = \exp[s(x,y)]$, yielding the illuminated-normalized enhanced image.

(d) Justify your selected approach with reasoning. Homomorphic filtering is uniquely suited for uneven illumination because standard spatial enhancement methods (like global histogram equalization) fail when

lighting varies spatially—equalizing an unevenly lit image simply brightens the glare and darkens the shadows further. Homomorphic filtering explicitly separates the illumination component $i(x,y)$ from the reflectance component $r(x,y)$ in the log-frequency domain, selectively dampening slow lighting variations while boosting high-frequency object details.

(e) Provide a clear and well-structured explanation. By taking the logarithm of the image, multiplicative lighting variations become additive. Passing the result through a specialized frequency filter suppresses low-frequency illumination trends ($\gamma_L < 1$) while boosting high-frequency reflectance details ($\gamma_H > 1$). Taking the inverse Fourier transform and exponentiating restores an image with flat, uniform background lighting where object details are perfectly clear across both formerly dark and bright regions.

3. Salt-and-Pepper Noise

Scenario Statement: A scanned document contains salt-and-pepper noise affecting readability.

(a) Explain the scenario and characteristics of this noise. In document image processing, salt-and-pepper noise (also known as impulse noise or data drop-out noise) appears as sparsely distributed, isolated pixels with corrupted intensity values. Specifically, 'pepper noise' presents as pure black pixels (intensity value 0 in 8-bit grey) scattered across light background areas, while 'salt noise' presents as pure white pixels (intensity value 255) scattered across dark character strokes. This noise is typically introduced during document scanning due to dust or scratches on the scanner glass, faulty CCD sensor elements, or corrupted signal transmission over noisy channels.

(b) Identify key issues impacting image quality.

- **Text Disruption & Character Fragmentation:** Black pepper noise pixels falling directly on character strokes create false gaps, severing thin letter lines (such as in letters 'e', 't', 'i', 'o') and rendering text unreadable.
- **Background Foreground Corruption:** White salt noise dots on clean background regions act as false positive foreground artifacts.
- **Optical Character Recognition (OCR) Failure:** Automated document digitization software (OCR) relies on clean connected-component analysis. Impulse noise fragments character contours, leading to high character misclassification rates.

(c) Apply an appropriate filtering technique. To remove salt-and-pepper noise effectively without destroying text characters, we apply a Non-Linear 3x3 Median Filter. The median filter slides a 3x3 spatial window over the image pixel-by-pixel, extracts the 9 surrounding intensity values, sorts them in numerical ascending order, and replaces the center pixel value with the statistical median value:

- **Neighborhood Extraction:** For pixel (x,y) , extract 3x3 neighborhood $W = \{P_1, P_2, P_3, P_4, P_5, P_6, P_7, P_8, P_9\}$.
- **Sort Operation:** $W_sorted = \text{Sort_Ascending}(W) = [I_min, \dots, I_median, \dots, I_max]$.
- **Replacement Rule:** $\text{Pixel_New}(x,y) = I_median = W_sorted[5]$ (the 5th value in the 9-element sorted array).

(d) Justify why this method is more suitable than alternatives. Linear smoothing filters (such as Mean or 3x3 Averaging filters) compute output as the weighted average of neighbors. When an impulse noise value (0 or 255) is averaged into a neighborhood, it pulls the average significantly, spreading the isolated noise dot into a blurry grey smudge that permanently ruins text legibility. In contrast, the Median Filter is a non-linear order-statistic filter. Because noisy pixels (0 or 255) represent extreme mathematical outliers, they fall at the absolute ends of the sorted array (index 1 or index 9) and are completely discarded when picking the middle median value (index 5). Thus, noise is 100% eliminated while crisp character edges remain completely unblurred.

(e) Present your response clearly and logically. By sliding a 3x3 median window across the scanned document, isolated impulse noise values (0 or 255) are filtered out because they occupy extreme rank positions in sorted order. The center pixel is replaced by the median intensity of surrounding valid document pixels, completely purging salt-and-pepper dots while preserving character geometry and restoring full text readability.

4. Blurred Edges

Scenario Statement: An image processed for noise removal appears overly smooth, causing edges to blur.

(a) Interpret the scenario and explain why edges are blurred. In image processing, sharp object boundaries, fine lines, and fine details correspond mathematically to high spatial frequencies (rapid intensity changes over short spatial distances). When low-pass spatial smoothing filters (such as Gaussian filters or large Box averaging masks) are applied for noise reduction, they function by attenuating high spatial frequencies indiscriminately. While this successfully smooths out noise spikes, it simultaneously suppresses high-frequency edge gradients, converting sharp step-function intensity transitions at boundaries into gradual, blurry intensity slopes.

(b) Identify key issues related to smoothing and edge loss.

- **Loss of Edge Crispness & Boundary Localization:** Object contours lose sharp definition, making exact physical boundary localization imprecise.
- **Erasement of Fine Structural Features:** Thin lines, sharp corners, junctions, and subtle surface textures are completely erased during heavy smoothing.
- **Degradation of Edge Detection Operators:** Gradient-based edge detection algorithms (such as Sobel, Prewitt, or Canny) fail to detect weak, smoothed intensity slopes, causing broken object outlines in downstream segmentation.

(c) Apply an appropriate technique to restore edges. To restore crisp edge boundaries without re-amplifying noise, we apply Unsharp Masking and High-Boost Filtering (or a second-derivative Laplacian Sharpening Operator). Unsharp masking isolates high-frequency detail components and adds them back proportionally into the image:

- **Step 1 (Smoothing):** Obtain a smoothed version of the image $f_{\text{smooth}}(x,y)$ using low-pass filtering.
- **Step 2 (Detail Mask Extraction):** Subtract the smoothed image from the original image to isolate the high-frequency detail mask: $g_{\text{mask}}(x,y) = f(x,y) - f_{\text{smooth}}(x,y)$.
- **Step 3 (High-Boost Addition):** Add a weighted portion of the detail mask back to the original image:
 $f_{\text{sharpened}}(x,y) = f(x,y) + k * g_{\text{mask}}(x,y)$, where $k \geq 1$ ($k=1$ gives standard unsharp masking; $k > 1$ gives high-boost filtering).

(d) Justify your decision with clear reasoning. Unsharp masking is selected because it selectively isolates high-frequency edge details in the detail mask $g_{\text{mask}}(x,y)$ and re-amplifies them along boundary transitions. By adjusting the weight factor k , we can control the exact degree of edge steepening. This steepens blurry intensity slopes back into sharp step transitions, restoring clear object outlines without distorting homogeneous background regions.

(e) Structure your explanation clearly. By subtracting the low-pass smoothed image from the original image, we extract an explicit high-frequency detail mask representing edges. Multiplying this mask by factor k and adding it back to the image steepens edge transition slopes along object boundaries, bringing back crisp edges and restoring structural sharpness across the image.

5. High-Frequency Noise

Scenario Statement: A satellite image contains fine-grained noise affecting feature visibility.

(a) Interpret the scenario and describe high-frequency noise. In remote sensing and satellite image analysis, fine-grained noise (such as sensor thermal noise, atmospheric speckle, or electronic interference during signal transmission) manifests as rapid, random intensity variations across short spatial distances. In the 2D Discrete Fourier Transform (DFT) frequency domain, these rapid spatial variations correspond to high spatial frequencies located far away from the origin (center DC component). This fine-grained noise degrades terrain feature visibility and corrupts land-cover classification accuracy.

(b) Identify the key issues affecting image clarity.

- **Masking of Subtle Terrain Features:** Fine speckle noise obscures small agricultural boundaries, narrow river streams, roads, and geological fault lines.
- **Spurious Edge Detections:** Gradient-based edge detectors detect thousands of false edges caused by noise spikes rather than real ground features.
- **Spectral Clustering & Classification Errors:** Automated multi-spectral land-use classification algorithms (e.g., k-means clustering) misclassify noisy pixels, producing fragmented classification maps.

(c) Apply an appropriate frequency domain filtering method. To suppress fine-grained high-frequency noise without introducing severe ringing artifacts, we apply a 2D Butterworth Low-Pass Filter (BLPF) in the frequency domain. The transfer function $H(u,v)$ of a Butterworth Low-Pass Filter of order n with cutoff frequency D_0 is defined as:

- **Butterworth LPF Transfer Function:** $H(u,v) = 1 / [1 + (D(u,v) / D_0)^{(2n)}]$
- **Distance Calculation:** $D(u,v) = \text{sqrt}[(u - M/2)^2 + (v - N/2)^2]$, representing distance from frequency origin $(M/2, N/2)$.
- **Frequency Filtering Pipeline:**
 1. Compute 2D FFT of image: $F(u,v) = \text{FFT2}\{ f(x,y) \}$
 2. Shift DC component to center: $F_{\text{shifted}}(u,v)$
 3. Multiply by Butterworth Filter: $G(u,v) = H(u,v) * F_{\text{shifted}}(u,v)$
 4. Inverse FFT to spatial domain: $f_{\text{enhanced}}(x,y) = \text{Real}(\text{IFFT2}\{ G(u,v) \})$

(d) Justify the choice of filter with reasoning. An Ideal Low-Pass Filter (ILPF) has a sharp, discontinuous cutoff at D_0 that causes severe 'ringing' artifacts (Gibbs phenomenon) across the image, creating fake ripple patterns that ruin satellite terrain interpretation. On the other hand, a Gaussian Low-Pass Filter (GLPF) provides smooth filtering but causes excessive blurring. The Butterworth Low-Pass Filter (BLPF of order $n=2$) provides the optimal trade-off: its smooth, continuous transition around cutoff frequency D_0 attenuates high-frequency noise effectively while completely eliminating ringing artifacts.

(e) Present your answer in a clear and organized manner. By transforming the satellite image into the 2D Fourier frequency domain, high-frequency speckle noise is concentrated in outer frequency regions. Multiplying by the 2D Butterworth Low-Pass Filter transfer function $H(u,v)$ smoothly attenuates frequencies beyond cutoff D_0 . Inverse Fourier transformation yields a clean, noise-free satellite image with smooth terrain regions and clear geological features.